

REVIEW ARTICLE

A Global Review of Monitoring Cropland Abandonment Using Remote Sensing: Temporal–Spatial Patterns, Causes, Ecological Effects, and Future Prospects

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Using remote sensing methodologies to uncover the temporal–spatial patterns of cropland abandonment (CA) offers substantial advantages at both macro scales and in real time. However, the current literature lacks a systematic review of specific typologies and methods regarding the application of remote sensing technology to CA monitoring. To address this knowledge gap, we systematically reviewed remote sensing-based methods for monitoring CA, its causes, and ecological effects. Our results show that the methods for monitoring abandoned cropland can be classified into 2 major categories: those based on image spectral features and those based on land cover temporal trajectories and vegetation phenology dynamics. Among the 8 subcategories, vegetation phenology and dynamic methods exhibit the highest average overall accuracy at $89.33\% \pm 3.37\%$. Remote sensing plays a crucial role in assessing the causes of CA, such as road density, spatial information of agricultural infrastructure, and the ecological effects resulting from abandonment, including food loss risks, carbon sequestration, wildfire risk, evapotranspiration, wilderness quality, biodiversity, and climate change. Further advancements are needed in classifying abandoned cropland types, observing fragmented and temporally unstable parcels, and assessing ecological effects across different scenarios. More importantly, we presented a trinity CA monitoring framework based on the cause–pattern–effect pillars, which offers a novel perspective for comprehensive research on CA. Overall, our work provided a systematic and insightful perspective for advancing remote sensing research on CA.

Introduction

Cropland abandonment (CA) is a widespread land cover change phenomenon globally [1,2]. Despite the inevitable future expansion of cropland to support global population growth, dietary changes, and efforts to eradicate hunger (SDG 1), CA and agricultural land expansion occur simultaneously [3,4]. Projections from the Shared Socio-Economic Pathways (SSP) scenario framework estimate that the total area of cropland will expand by 25 to 226 Mha over the next 3 decades, with variation across the most sustainable to the least sustainable scenarios [2]. Nevertheless, United Nations reports indicate that the world

loses 12 Mha of productive land every year, with far-reaching consequences [5]. The area of abandoned cropland is expected to increase further in the next decade [6]. Therefore, obtaining accurate temporal–spatial information on CA is essential for in-depth analysis of its causes, ecological effects, and the development of sustainable land management strategies [7].

The remote sensing-based CA monitoring program provides efficient, convenient, and wall-to-wall datasets to capture abandoned land with spatially explicit information [8,9]. The common paradigm for identifying abandoned cropland using remote sensing typically distinguishes it from permanent farmland based on the physicochemical vegetation spectral characteristics

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and landscape patterns. The spectral reflectance characteristics of abandoned land substantially differ from those of long-tilled farmland, particularly at the vegetation level [10]. From a dynamic perspective, the vegetation on abandoned land typically undergoes secondary succession, with vegetation cover increasing annually [11]. In contrast, the time cycle of cropland shows substantial seasonal fluctuations in vegetation levels [12]. Vegetation cover indices, such as the Normalized Difference Vegetation Index (NDVI) and Normalized Burn Ratio [13], provide robust pseudocolor spectra for CA monitoring within vegetation phenology and the crop life cycle [14,15]. Geographically, abandoned land is often characterized by rougher and more irregular surface textures due to the lack of human management, whereas cultivated farmland exhibits regular crop spacing [16]. Characterized by unique geophysical features, such as high slopes, low soil quality, and fine-grained textures, mountainous regions frequently experience CA [17,18]. From a landscape perspective, abandoned land exhibits high fragmentation and heterogeneity [19]. Agricultural land that is distant from paved roads or isolated islands surrounded by natural forests is more susceptible to abandonment.

With the advancement of remote sensing technology, high temporal-spatial resolution images are now available to capture surface physical changes, providing precise options for monitoring and analyzing CA phenomena. Landsat TM and Sentinel satellites, with their various sensors, provide fine spatial resolution images widely employed in identifying CA [13,20]. The Moderate Resolution Imaging Spectroradiometer (MODIS) is often utilized in large-scale CA surveys, such as those in Europe and China [21,22]. It offers high-frequency access and periodicity, making it suitable for monitoring CA. Nowadays, sub-meter-level image resolution, such as that from GF-series and aerial imagery, provides finer pixel-level information for monitoring abandoned cropland [23,24]. The light detection and ranging (LiDAR) technology can penetrate vegetation layers to capture detailed height information about the ground surface and vegetation canopy, helping researchers monitor the successional dynamics of abandoned land, including changes in vegetation cover and surface biomass [25].

Remote sensing cloud computing platforms offer efficient and flexible solutions through distributed computing and big data technologies, making large-scale CA monitoring a reality [26]. Google Earth Engine (GEE) provides robust technical support for large-scale CA monitoring. Advances in algorithms and computational power have facilitated the widespread adoption of classical classification methods such as support vector machines (SVMs) [27], random forest [28], and decision tree algorithms, substantially improving performance. These advancements not only improve monitoring accuracy but also capture small differences in the spectral and textural features of CA [29]. Notably, machine learning and deep learning algorithms have introduced diverse methods for identifying CA, with neural network models demonstrating high-precision performance in the rapid processing of remote sensing images for CA monitoring [27].

CA is a complex phenomenon driven by multiple factors, with a wide range of ecological effects [23]. Remote sensing data and methodologies play a crucial yet often overlooked role in analyzing the causes and ecological impacts of CA. Environmental drivers of CA processes, such as elevation, slope, and soil erosion, are directly or indirectly observable through satellite imagery. Socioeconomic factors also are considered the main drivers of CA [30]. Geospatial information

technology based on remote sensing can help decipher the potential socioeconomic factors behind CA. For example, factors such as the distance from arable land to roads, accessibility to water resources, economic development (proxied by nighttime lighting), orchard establishment, and policies related to the return of farmland to forests can be analyzed using remote sensing [31]. Additionally, remote sensing-based investigations of the ecological effects following CA, such as carbon sequestration, soil and water conservation, provisioning services, and wildfire risks, are also vital [32,33]. Model simulations using spatially explicit methods can predict potential ecosystem services and functions resulting from CA [2].

Despite advances in remote sensing methods for monitoring CA, large uncertainties and quantification gaps remain due to variations in methods and model outputs [34]. Even within the same region, quantitative temporal-spatial information on abandoned land may be inconsistently influenced by image resolution, sample size, and external methodologies [35]. Differences in the definition of CA also influence discrepancies in the results [36]. Furthermore, remote sensing methodologies must be selectively applied for monitoring CA at different spatial scales in the study area, considering external factors such as computational power and the estimated average patch size of abandoned land. There is a lack of a comprehensive overview of remote sensing methodologies for CA monitoring. Additionally, its causes, temporal-spatial dynamics, and ecological effects need to be thoroughly explored.

To fill this knowledge gap, this paper reviews the dynamics of global research on CA, highlighting its temporal-spatial monitoring, causes, ecological effects, and future prospects. Specifically, we searched peer-reviewed papers, reports, books, and reviews in the Web of Science (webofscience.com) bibliographic database up to 2024 May 23 using the following keywords: (agricultural OR arable OR cropland abandonment) AND (abandoned cropland OR remote sensing OR mapping) AND (driving force OR causes OR ecological effects). By manually examining the titles and abstracts, we broadly classified the articles into 3 dimensions: land abandonment mapping, causes, and ecological effects using Zotero (zotero.org) literature management software. A total of 254 articles were ultimately included in this review, comprising 105 on temporal-spatial monitoring, 61 on causes, and 88 on ecological effects (Tables S1 to S3). Additionally, we discussed the challenges and uncertainties of current research related to CA and provided future prospects. Overall, we synthesized the literature on CA and remote sensing methodologies and aim to advance CA research toward more accepted and consistent methodological and scientific practices.

The Concept of CA and Its Current Research Status

The concept of CA

Differences in definitions offer an opportunity to assess CA from various perspectives, enabling the development of in-depth evidence and a broader understanding of the extent and processes of land abandonment at multiple levels [37]. For comparison, a clear and universally accepted definition of CA is essential for further analysis and consistent research. Currently, CA is broadly defined; its definitions need to account for factors such as time span, land management, land cover changes, and other dimensions [1].

In the time-span dimension, CA refers to cropland that has been left unused or underutilized for a certain period. In 1995,

the Food and Agriculture Organization (FAO) defined CA as the cessation of farming activities for at least 2 to 5 years (Table 1), during which no evidence of cultivation is found and natural vegetation is restored [18]. The American Farm Bureau defines abandonment as occurring when unimproved farmland has not been used for agricultural purposes at least once in the preceding 5 years (Table 1). Regarding this definition, the inactivity period associated with the transition from cropland to abandoned land varies across different studies [31]. Considering the influence of land management policies, some studies in China define CA as farmland left unused for at least 1 year or even a season. Additionally, inactivity periods of at least 2 years [38], 3 years [39], and 5 years [2,40] have been used as basic conditions for recognizing CA. There is no consensus on the transition period, which is likely closely linked to agricultural management policies and practices [41].

From the perspective of agricultural land management, CA represents a gradual decline in agricultural intensity, eventually leading to underutilization or complete cessation of farming activities, with no intention of replanting [37]. The European Union's Institute for Environmental Policy (IEEP) classifies arable land abandonment into 3 categories: complete abandonment, semi-abandonment, and excessive abandonment, underscoring their significance for agriculture and ecosystems in guiding policy and land management practices (Table 1). This definition underscores the significance of CA for agriculture and ecosystems, providing guidance for policy development and land management practices. From an agronomic perspective, abandonment is a process marked by a gradual reduction in land-use intensity, ultimately resulting in the cessation of agronomic inputs and the complete termination of farming activities [37].

In recent years, CA has been defined in terms of land-use type and intensity, reflecting the transformation of agricultural land into early-successional natural vegetation [1]. Widely recognized as a substantial land cover change, CA involves the permanent conversion of agricultural land to unmanaged grasslands, fallow land, successional scrub, or young forests [42]. This perspective integrates concepts from remote sensing-based land cover change and trajectory detection into CA monitoring, enabling studies in various regions, including global contexts

[2], conflict zones [43], and mountainous regions [44], among others.

CA is a dynamic process that describes the transition from active cultivation to cessation of agricultural production. In contrast, abandoned land refers to a static state where land or farmland has completely ceased agricultural activity. Clarifying the concept of CA requires distinguishing it from related terms (Fig. 1). Fallowing is an agricultural management practice that involves temporarily ceasing plowing to restore soil fertility, reduce pests and diseases, and enhance agricultural sustainability (Fig. 1). Shifting cultivation is a traditional practice in which land is cultivated for a period before being abandoned for another plot, allowing the original land to naturally replenish its fertility (Fig. 1). Cropland retirement refers to government or agricultural policies that actively encourage the conversion of arable land to nonfarming status [45]. These policies typically aim to promote environmental protection, soil and water conservation, and ecological restoration, or reduce agricultural overproduction (Fig. 1).

Typology of CA

Through a comprehensive review of CA types, we found that CA can be categorized into 6 different types, based on their time span, drivers, and manifestations, respectively (Fig. 2). Regarding the time span of abandonment, CA can be classified as year-round abandonment and seasonal abandonment [46]. Year-round abandonment refers to the long-term or permanent cessation of cultivation on land, allowing it to naturally recover and no longer be used for agricultural production. Seasonal abandonment, on the other hand, involves leaving the land fallow during specific seasons or cycles, allowing it to rest before being ploughed again. Both types of abandonment are measures aimed at protecting the land, maintaining soil fertility, and regulating ecological balance, but they differ in implementation timing and duration. Seasonal abandonment rates have been reported to exceed those of perennial abandonment, although relatively little information is available on seasonal abandonment [46].

Based on the driver differences of CA, CA can also be categorized into active and passive abandonment. Active abandonment

Table 1. Concept of CA as defined by authoritative governments or institutions

Organization	Concepts
World Food Organization (FAO)	The cessation of farming activities for at least 2–5 years, during which no evidence of cultivation is found and natural vegetation is restored [140].
EU Institute for Environmental Policy (IEEP)	Complete abandonment, i.e., arable land that has been entirely abandoned for management; semi-abandonment, i.e., arable land with minimal management that has not yet been formally abandoned; and transitional abandonment, i.e., arable land that has undergone a change in land use or land-use type [54].
The U.S. Department of Agriculture (USDA)	Abandonment occurs when previously converted cropland has not been used for, or in support of, agricultural purposes at least once during the immediately preceding 5 years [141].
Department of Natural Resources of Hebei Province, China	Cultivated land that is idle or underutilized due to voluntary abandonment by landowners is influenced by a variety of factors, including natural, social, economic, and policy influences [142].

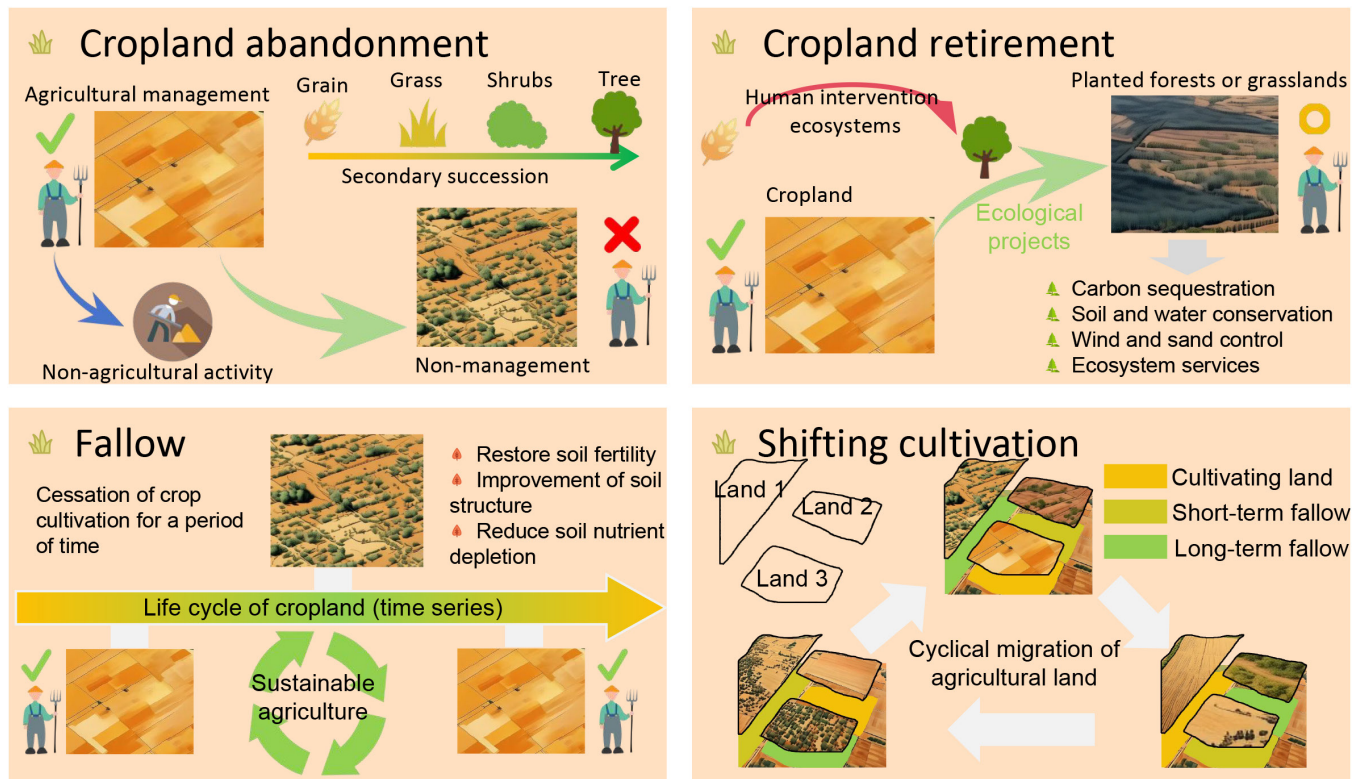


Fig. 1. Visualization map of CA and its relation to agricultural activities.

occurs when farmers consciously cease land management, often for purposes such as land restoration, environmental protection, or agricultural restructuring. In China, for example, various ecological restoration projects have aimed to convert low-productivity arable land into ecologically valuable areas to enhance carbon sequestration and improve soil and water conservation [47]. Passive abandonment, in contrast, occurs when farmers are unable to continue cultivating and managing their land due to external factors such as natural disasters, armed conflict, economic hardship, or policy constraints, leading them to abandon agricultural use.

Additionally, the concepts of explicit and invisible abandonment have emerged in recent years (Fig. 2). Explicit abandonment refers to the deliberate cessation of cultivation for production purposes, often resulting in visible changes in land cover, such as the growth of pioneer plants. As contrasting concepts based on their manifestations, explicit abandonment is frequently accompanied by observable land cover changes. In contrast, invisible abandonment occurs when land continues to be cultivated but with low efficiency or productivity due to factors such as soil infertility. Both explicit and invisible abandonment indicate inefficient land use, but while explicit abandonment involves visible changes in land cover, invisible abandonment is driven by underlying, less apparent factors. Therefore, studies related to CA using remote sensing technology primarily focus on explicit abandonment.

Current status of research on CA and its global pattern

CA is a widespread and enduring phenomenon of land-use change globally [13] affecting regions around the world to varying degrees [2]. We analyzed peer-reviewed papers on remote

sensing-based monitoring of CA and spatially visualize their quantitative information, including location, CA rates, overall mapping accuracy, and frequency of satellite use (Fig. 3 and Table S1).

Europe and Asia are the most frequently studied regions concerning CA, characterized by high CA rates (Fig. 3A). Additionally, countries in Latin America, Central Asia, and Southeast Asia are also experiencing CA. Eastern and Central Europe, as well as the Mediterranean rim, exhibit the highest prevalence of CA [48]. In China, CA is most pronounced in economically underdeveloped areas, such as mountainous regions in the southwest, Karst areas, arid and semi-arid zones, and the eastern plains. Bibliometric analyses and spatial results indicate that the peak CA rate in developed countries, such as the United States and the European Union, mostly occurred before 2000, whereas in China, it has been most notable between 2010 and 2020 (Fig. 3A) [49]. Notably, Africa is projected to become a focal area for future CA, following Europe and Asia [2].

The collapse of socialism precipitated profound socioeconomic and institutional changes, altering the land system and leading to widespread farm abandonment [50]. CA is especially severe in marginal mountainous areas where access is difficult and soil quality is poor [51]. Eastern European countries show an average CA rate of $27.16\% \pm 19.55\%$ (standard deviation), the highest among the 5 typical regions studied (Fig. 3D). Additionally, investigations indicate an average war-induced CA rate of $16.97\% \pm 4.42\%$, including conflicts in South Sudan, Russia, and Ukraine [52,53]. War-related destruction often involves the dismantling of infrastructure, such as roads, irrigation systems, and agricultural equipment, forcing farmers to abandon their land and triggering large-scale population displacement, including refugee

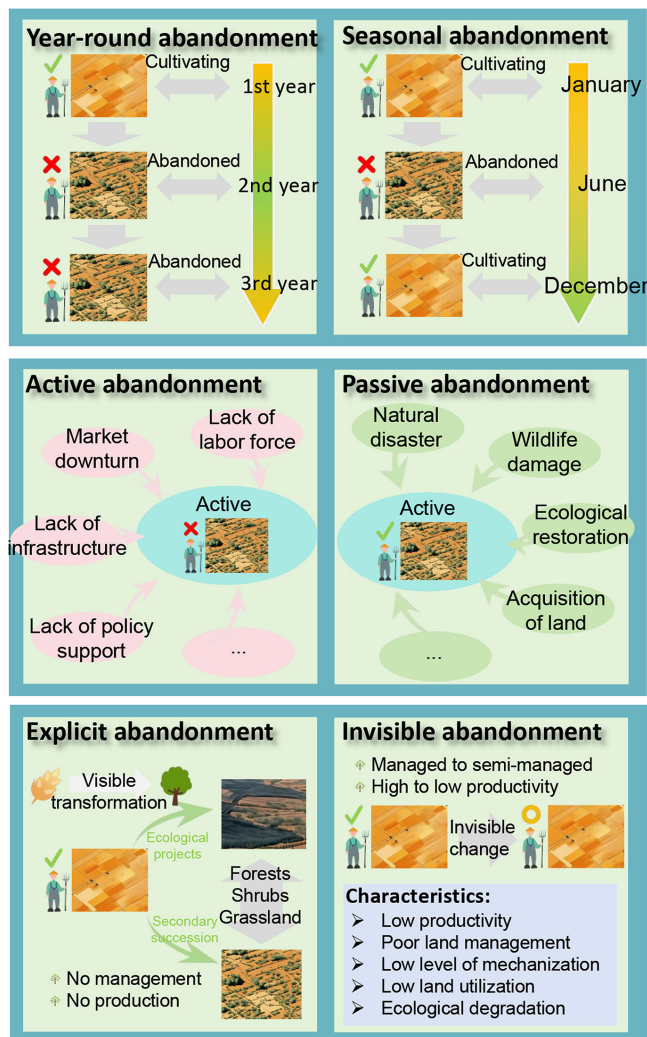


Fig. 2. Visualization map concerning the 6 types of CA and their differences.

flows, with lasting impacts on agricultural labor and post-war recovery efforts.

Bibliometric analysis revealed that more than one-third (39.47%) of studies rely on Landsat imagery. MODIS satellite data (13.16%), prized for their broad coverage and rapid revisit cycles, were frequently used for large-scale identification of fallow lands based on phenological characteristics (Fig. 3B). Some studies combined multiple satellite datasets (14.47%), often leveraging Landsat's high resolution and MODIS's frequent revisit capabilities, yielding robust results in identifying abandoned lands. Sentinel images (7.89%), Systeme Probatoire d'Observation de la Terre (SPOT) (3.95%), Quickbird (3.95%), and GF-series images were also commonly employed for identifying smaller-scale abandonment (Fig. 3B). Overall, the size of the study area and the distribution of abandoned lands dictate the most suitable satellite-based methods for monitoring.

Remote Sensing Methodologies for Monitoring Abandoned Cropland

Image classification and spectral features method Visual interpretation

Visual interpretation is a traditional method for extracting abandoned cropland by directly observing remote sensing

images using the human visual system. This methodology allows for comprehensive analysis, comparison, and judgment by identifying feature characteristics such as shape, size, color, texture, and spatial location on the image, and then uses contour lines to delineate the abandonment status [54].

Aerial imagery is the primary data source for visual interpretation to obtain spatial information on early abandoned cropland [53,55]. Using this method, Arnaez et al. [60] identified 6 types of abandoned land in the Camero Viejo region over the last 50 years, including abandoned terraces and abandoned valley cropland. Chen et al. [56] revealed the spatially explicit pattern of CA-induced abandoned land due to the construction of large reservoirs in China. Visual interpretation of abandoned land maps is often combined with field surveys [57]. For instance, Lieskovský et al. [58] used national land survey data to depict the secondary succession dynamics from cropland to natural forests in the mountainous areas of Switzerland. This method typically requires higher-resolution imagery, often sourced from aerial photographs or sub-meter pixel units, such as QuickBird and SPOT satellites [10].

Visual interpretation has the advantages of simplicity, higher accuracy, and user flexibility (Table 2). However, this approach is generally more suitable for local government-driven landscapes and rural areas, as it is limited by low levels of automation and large labor and time costs [59]. Additionally, the method heavily relies on the experience and knowledge of the interpreter, making it challenging to generalize on larger scales.

Supervisory classification

Supervised classification is a widely applied remote sensing technique for identifying abandoned land (Table 3). Remote sensing satellites capture valuable spectral information across multiple bands, providing distinct spectral characteristics of CA. By training the model on samples of abandoned land, a classification algorithm can be developed to distinguish between abandoned and nonabandoned land.

In practice, early CA monitoring relied on traditional, simple algorithmic models based on probability and mathematics, which are also commonly used in land classification studies. For instance, Lotfi et al. [60] trained a maximum likelihood classifier using multitemporal Landsat remotely sensed imagery to identify abandoned land through various false-color composite bands. Similarly, Witmer and O'Loughlin [61] calculated localized CA in Bosnia and Herzegovina using the minimum distance model.

With the development of machine learning technology, supervised classification methods have become more advanced, capable of handling high-dimensional and nonlinear relational data. This has made them the most commonly used methods for detecting abandoned cropland [61,62]. For example, Baumann et al. [63] mapped post-socialist abandoned farmland using SVM methods. Yoon and Kim [27] detected abandoned farmland using harmonic analysis and machine learning. Karim et al. [64] improved farmland abandonment detection using deep learning vision transformer and multiple vegetation indices, achieving better accuracy with high-resolution GF-6 remote sensing imagery. Lesiv et al. [65] used a Bayesian classification algorithm to calculate the extent of CA in post-socialism. Skakun et al. [66] identified farmland damage caused by military conflict in south-eastern Ukraine by combining remotely sensed imagery with a multilayer perceptron approach. The combination of multiple spectral bands has proven to be exponentially more effective in

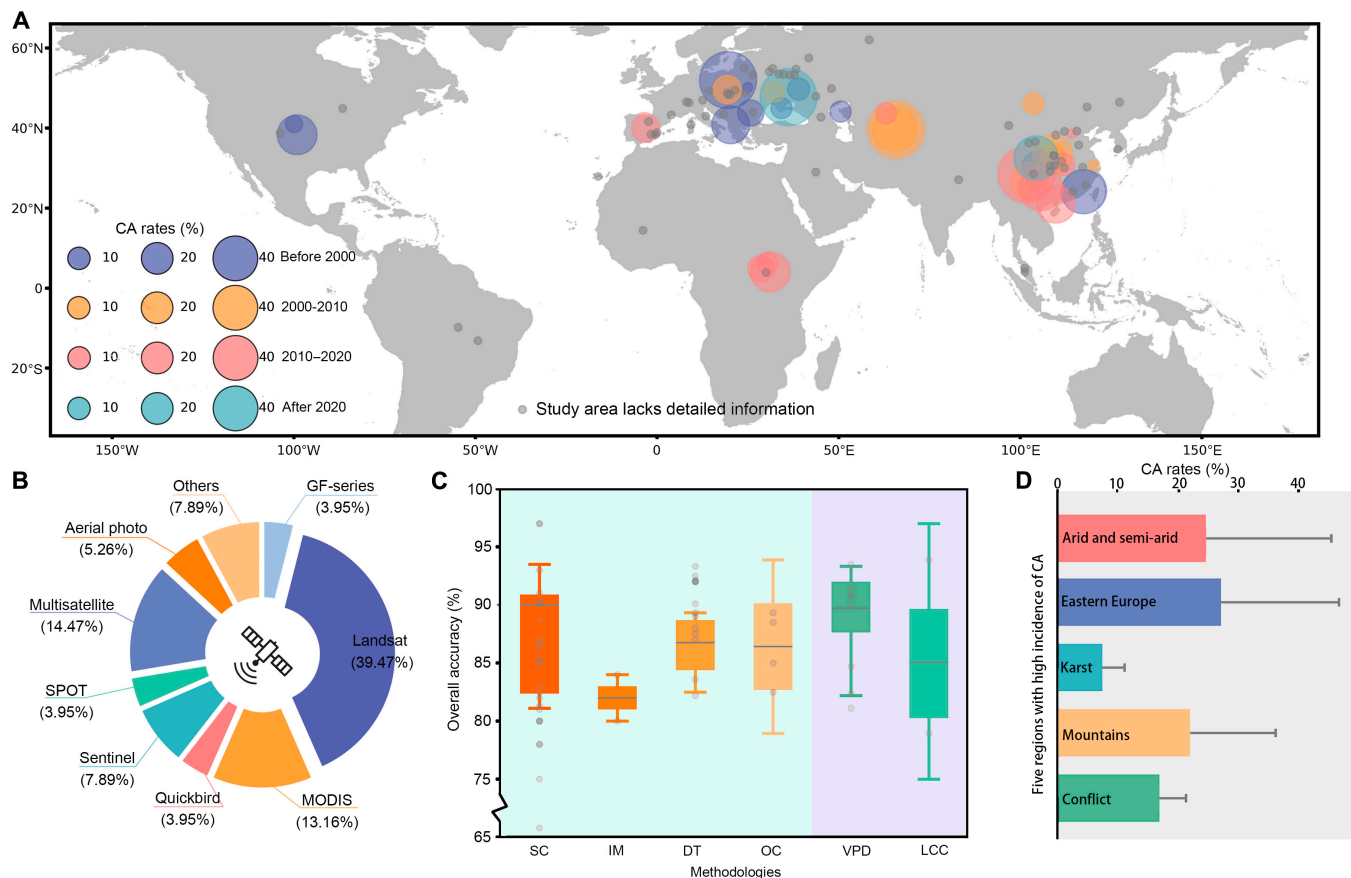


Fig. 3. Global distribution of remote sensing-based CA studies. (A) Global temporal-spatial distribution of CA study areas and their corresponding CA rates. (B) Share of satellites used for cropland abandonment monitoring. (C) Overall accuracy of abandoned land classification by different methods. (D) CA rate in 5 typical geographic regions. SC, supervised classification; IM, integrated methods; DT, decision tree; OC, object-oriented; VPD, vegetation phenology and dynamic; LCC, land cover change. We collated the study areas from 112 peer-reviewed papers and converted them to geographic coordinates using Google Maps (maps.google.com), except for 3 global-scale studies. In (C), the 2 background colors distinguish the 2 main classification methodologies (Table 2). In (D), we qualitatively differentiate 5 high-frequency abandonment areas from previous studies. Eastern Europe is included because the region has experienced substantial changes in agricultural land management systems.)

abandonment monitoring studies. Researchers often use seasonal NDVI indicators, along with climatic information, such as NDVI amplitude and frequency [67], as input data for supervised classification, which can improve the accuracy of identifying abandoned land.

Although supervised classification methods have made large progress, they require a large number of training samples, which is a time-consuming and labor-intensive process. The quality of these samples directly affects the classification results (Table 2). If the classifier is too complex or the training samples are unbalanced, the results can be compromised. In such cases, a migration method based on abandoned land samples is often adopted [45]. Additionally, current research on CA monitoring faces several challenges, such as differentiating between similar land classes (e.g., bare ground, grassland, cultivated land, and shrubs), dealing with noise from noncultivated land changes, addressing the lack of temporal and spatial resolution in the data, and overcoming interference from cloud cover and terrain.

Decision tree

Based on differences in image spectral features, another method involves constructing decision trees using feature information to identify abandoned land, either directly or indirectly (Table 2).

The identification of abandoned land using decision tree algorithms can be classified into 2 types.

On one hand, decision tree algorithms can be constructed to directly distinguish between abandoned and nonabandoned land. By constructing multiple decision trees, Morell-Monzó et al. [20] proposed an optimal identification algorithm for abandoned citrus orchards. Grădinaru et al. [10] tested multiple models to determine the most optimal model for mapping abandoned land during the vegetation growing season using classification and regression trees.

On the other hand, decision trees can categorize land use, and by comparing pixel-level land-use changes, abandoned land can be indirectly identified [22]. For example, Xiao et al. [66] prepared annual land-use maps from 1990 to 2017 using the Classification and Regression Tree (CART) classification method based on Landsat and HJ-1A image data. This proved crucial for mapping abandoned land. Similarly, Günthert et al. [81] generated consistent samples from 2000 to 2020 for classification purposes and identified abandoned farmland pixels based on farmland conversion.

Decision trees are an efficient method for mapping abandoned land, capable of reducing time and labor by nearly half [10]. The structure of the decision tree is intuitive, resembling human decision-making, making it easy to understand, interpret,

Table 2. The list of abandoned cropland mapping methodology and its principles, characteristics, and applicable scales

Overall methods	Specific methods	Principles	Strength	Weaknesses	Applicable scales	References
Image classification and spectral features method	Visual interpretation	Abandoned land identification by experts	High accuracy; vector data; user-flexibility [53]	High subjective; high workforces and time costs	Landscape to rural scale	[58,57]
	Supervised classification	Training classification models with abandoned land samples	User-simple; automated	Subjective selection of samples	Landscape to regional scale	[63,143]
	Decision tree	Abandoned land classification based on conditional splitting	Cost-effective; logically sound and interpretable	Vulnerable to spectral interference	Regional to national scales	[22,68]
	Object-oriented classification	Based on physical attributes	Considers both texture and geometric information	Segmentation threshold; requires high computational power	Landscape to rural scale	[9,59]
	Integrated methods	Integration of multiple methods	Combines the strengths of multiple methods to improve both accuracy and fault tolerance	Emphasis on opaque models	Regional to national scales	[67,76]
Time-dynamic and trajectory methods	Dual time point land cover changes	Overlay of cropland data from 2 periods	User-friendly and efficient	Greatly affected by land cover data	Regional and national scales	[43,66]
	Multi-time point land cover trajectory	Moving window approach with land cover pixels	Year-to-year; easily understood by users	Dependent on the land use/cover classification accuracy [84]	Regional to global scales	[106]
	Vegetation phenology and dynamic	Identifying differences in vegetation phenology	High accuracy and sensitivity; adaptable to diverse scenarios	Noise from noncultivated land; requires time-resolved images	Landscape to national scale	[84,94]

and present results. Decision trees handle nonlinear relationships in data very well. However, they also have disadvantages, such as overfitting and sensitivity to data variability (Table 2).

Object-oriented classification

In very-high-resolution images, abandoned farmland often exhibits a distinctly chaotic and disorganized texture. Unlike the orderly patterns of cultivated farmland and orchards, abandoned land lacks regular cultivation and management, typically being covered by grasses and shrubs [69]. To address this, an object-oriented classification method that synthesizes spectral, textural, and spatial information improves the accuracy of identifying abandoned land (Table 2).

To identify fine-grained and marginalized areas of abandonment, an object-oriented approach segments image data into multiple objects [6]. By extracting various visual textures from

the target region, deep learning techniques can be efficiently trained and applied. Using this method, Wang et al. [6] identified abandoned farmland using SPOT-5 high-resolution images. By introducing a neural network with a texture computation module, Shen et al. [59] conducted experimental work on mapping abandoned farmland in very-high-resolution images. Koleccka and Kozak [70] proposed wall-to-wall, parcel-level CA mapping in the Polish Carpathians. Recently, farmland segmentation frameworks integrating advanced semantic and textural features have emerged, such as the pixel attention contrastive network [71], gray level co-occurrence matrix [72], and pyramid scene parsing network–statistical texture learning [59].

Compared to traditional algorithms, using cultivated land parcels as the detection unit and applying segmentation techniques can effectively reduce errors [73]. Improvements in computer technology have facilitated the expansion of CA detection

Table 3. List of the 5 main supervised classification methods used for mapping of abandoned land

	Methods	Study area	Image sources	Overall accuracy	CAR
Supervisory classification	Maximum likelihood	Central Iran [60]	Cloudless Landsat images	82.36%	More than 20%
	VSM	Western Ukraine (Ivano-Frankivska, Lvivska, and Zakarpatska) [63]	Landsat TM images and Quickbird	93.5%	56%
		The border triangle of Poland, Slovakia, and Ukraine in the Carpathian mountain [143]	Landsat TM/ETM+ images	90.9%	16.1%
		Across Central and Eastern Europe and the Balkan Peninsula [144]	MODIS time-series satellite data	50.7%	16.8%
		Gwangyang City in South Korea [27]	Sentinel-2A	90.72%	-
	Random forest	Europe [145]	MODIS vegetation indices	90.10%	18.0%
		Oliva, in Spain [20]	Sentinel-2 and aerial photographs	77.10%	-
	Neural network	Mianchi County, China [64]	GF-6 imagery	91.5%	2.2%
		The Donetsk and Luhansk regions [66]	Landsat and Sentinel-2	84.7%	22%

units [74]. This methodology has also been implemented in countries such as Ukraine [75] and Malaysia [74]. However, the limitations of this method include the difficulty of selecting an appropriate image segmentation threshold and the risk of over-fitting when too many constraint rules are applied.

Integrated methods

Integrated method classification involves the fusion of two or more techniques to identify abandoned land (Table 2). On the one hand, this approach imposes additional constraints on abandoned land extraction. On the other hand, combining different methods can enhance overall classification accuracy and fault tolerance. De Castro et al. [76] investigated the temporal-spatial distribution of sugarcane land abandonment in Rio de Janeiro, Brazil, by integrating LandTrendR with multitemporal mapping. To monitor land abandonment, Löw et al. [67] used a probabilistic model that combines random forests and SVMs.

The integration of these high-precision methods yields complementary results and addresses challenges through decision fusion. Combining two or more methods substantially improves classification accuracy compared to using a single method alone (Table 2).

Time-dynamic and trajectory methods

Using time stack data, the CA process can be detected based on crop spectral characteristics at different observation time points by recognizing changes in feature curves. This method, known as the time-based dynamic and trajectory detection algorithm, is a primary approach for extracting information about abandoned cropland [77]. We broadly categorize these methods into 3 types: dual time points land-use change, multi-time point land

cover trajectory, and vegetation phenology and dynamics methods (Fig. 4).

Dual time point land cover changes

CA is one of the most widespread land change phenomena in the world [78]. After CA, land is no longer cultivated and managed, allowing vegetation and ecosystems to recover naturally through ecological succession and rewilding [79]. The surface of abandoned land often transitions into grassland or shrubland. By overlaying 2 pixel-level land cover maps, the temporal-spatial patterns of abandoned land can be accurately identified [80].

Remote sensing imagery from 2 periods—before and after CA—is crucial for this method. A dichotomous classification is applied to obtain information on cultivated and noncultivated land. The conversion of cropland to noncropland is identified as abandoned land (Table 2). This method is generally used to reveal war- and contingency-induced CA. For example, Olsen et al. [43] used Sentinel imagery to reveal the reduction of arable land from 2016 to 2018 before and after the war in South Sudan. Skakun et al. [66] found that the abandonment of CA in southeastern Ukraine has continued since the military conflict began in 2014, with socioeconomic impacts on the Donetsk and Luhansk regions. Nguyen et al. [81] compared Landsat imagery from around 1990 (before the collapse of the Soviet Union) and around 2015 (current situation) using supervised classification to estimate the extent and spatial distribution of abandoned farmland.

This method is user-friendly, fast, and efficient, with unique advantages in areas dominated by subsistence and smallholder farming systems [43]. However, the challenge lies in robustly

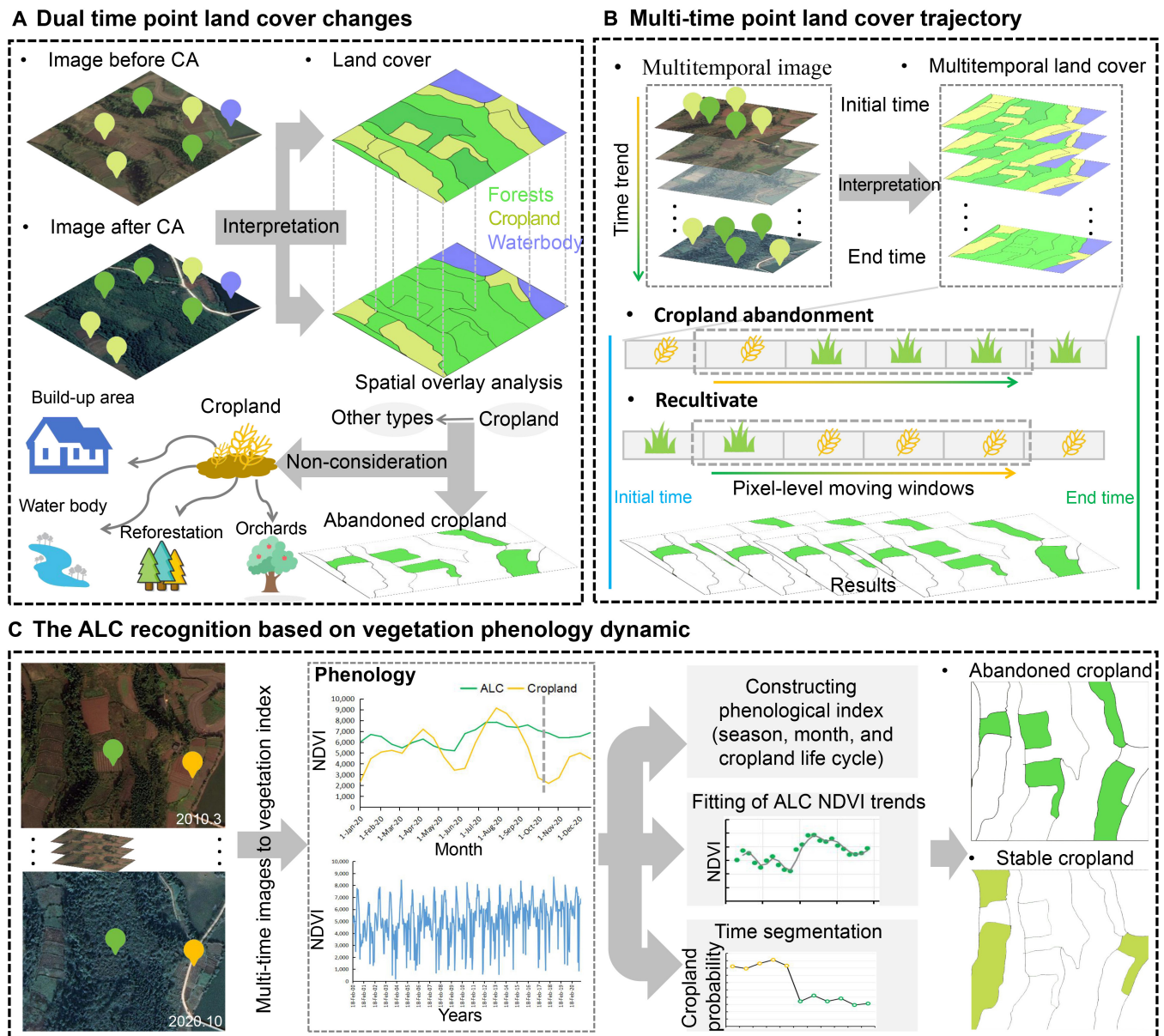


Fig. 4. The conceptual map for monitoring CA based on time dynamics and trajectory methods. (A) Dual time point land cover changes. (B) Multi-time point land cover trajectory. (C) Vegetation phenology and dynamics. Here, a case area (30°53'N and 105°51'E) and its phenology dynamic were shown using Landsat sensor.

demonstrating the stability and permanence of abandoned land using only 2-period land status data (Table 2). This method can also be affected by uncertainties in cropland fallow, compaction, and land cover data accuracy [65]. Additionally, rapid changes related to flooding in riparian areas may render the low signal-to-noise ratio of direct change detection unsuitable for this approach.

Multi-time point land cover trajectory

As agricultural land can remain unused for years, multitemporal or time-series satellite data are often needed to identify these areas. Based on the definition of CA proposed by the FAO, the methodology of land-use change trajectory has been widely used in recent years [82]. This method enables rapid monitoring of annually abandoned land with some temporal-spatial stability [83]. Nearly half of the articles chose the land change trajectory method to identify abandoned land (Table S1).

Direct change monitoring involves constructing a trajectory guideline using land-use changes at three or more time points [83]. Dara et al. [784] developed a trajectory-based change detection method to identify cropland wastelands between 1988 and 2013 and re-tillage between 1991 and 2013. Du et al. [45] mapped agricultural wastelands and retirement in North China based on temporal consistency checking, which is applicable to arid areas with slow secondary succession. Witmer [85] assessed the effectiveness of 4 change detection methods in detecting abandoned agricultural land. This method is suitable for capturing the spatial distribution of annually abandoned land and can be widely applied on a global scale [86]. It is also frequently used to monitor land reclamation [87].

In addition to hard classification, some studies have improved the time-trajectory-based approach by applying soft classification. Specifically, change detection algorithms utilize the time profile of the classification probabilities of cropland and

noncropland [84]. On the GEE platform, the LandTrendr [88] and Continuous Change Detection and Classification algorithm [89] can perform this process. These methods can detect sudden breaks and continuous trends, but there is a probability of CA if the farmland is spectrally highly variable.

Overall, the temporal trajectory detection method can attenuate the effects of negative factors such as atmospheric conditions, sensor differences, seasonal variations, and changes in the ground environment on different simultaneous images (Table 2). However, the key to this method is the accuracy of land classification. Since there are inherent errors in image classification, this can lead to a loss of accuracy in the change detection results and increased uncertainty in the outcomes.

Vegetation phenology and dynamic

Using single-season remotely sensed imagery, existing methods find it difficult to distinguish abandoned cropland with early successional regenerating grasses and shrubs from actively managed cropland [79]. Therefore, methods based on multi-temporal and multiseasonal imagery can detect phenological dynamics, which help distinguish crops from herbaceous plants [90].

The NDVI is a commonly employed vegetation spectral index for climatological identification of abandoned cropland [91]. Prishchepov et al. [92] gathered 6 near-annual cloud-free Landsat images from the “spring”, “summer”, and “fall” agricultural seasons to investigate the temporal-spatial patterns of abandoned agricultural land. Boudinaud and Orenstein [93] combined the maximum NDVI values at the beginning, middle, and end of the growing season to select cropland for each year of interest. Yin et al. [83] aggregated Landsat TM/ETM+ and Operational Land Imager time series for all years during 1984 to 2016 into annual NDVI composite maps to detect long-term gradual and short-term drastic changes in agricultural land use. Alcantara et al. [94] used a 16-day MODIS NDVI composite to analyze phenological indicators and separately map abandoned agriculture in Europe. Minghua et al. [11] extracted abandoned farmland based on changes in crop life cycles.

Bibliometric analysis has shown that monitoring abandoned land based on the vegetation phenology method achieves the highest average accuracy with 89.33% (Fig. 3). Vegetation index change detection identifies the difference in growth between crops and natural vegetation, suppresses atmospheric and environmental noise, requires low temporal resolution data, has low data redundancy, and has high fault tolerance (Table 2). Nonabandoned sites tend to have cyclical annual NDVI changes, while abandoned sites with natural vegetation succession tend to show increasing NDVI trends [93]. For monitoring abandoned land with a short time span, this method is ineffective in arid zones [45]. Additionally, in coarse-resolution imagery, the NDVI time-series characteristics of abandoned farmland may be confused with those of other land cover types.

Revealing the Causes of CA from a Remote Sensing Perspective

The phenomenon of CA is driven by multiple causes and factors [30], with the complexity of their combined effects resulting in varied responses across different regions [63]. Additionally, the causes of CA are often more relevant at the local scale [91,92]. Investigating the dominant factors of CA in different regions requires a broader scale and a top-down spatial analysis approach

[9]. In general, the causes of CA can be categorized into geographical and socioeconomic aspects [48].

CA is a spatially explicit phenomenon driven by land marginalization [95]. Remote sensing is a valuable tool for investigating the causes of CA, providing fast, efficient, and large-based information. Almost all large-scale CA studies involve raster-scale datasets derived from remote sensing, either directly or indirectly, as driver inputs. Remote sensing techniques identify landscape patterns of arable land, such as patch size, fragmentation, and aggregation [96]. For instance, Nguyen et al. [81] resampled all layers, including socioeconomic factors, to a 30-m resolution to match land use/land cover maps derived from Landsat satellite imagery. Prishchepov et al. [50] combined Landsat TM/ETM+ footprints of abandoned cropland maps with socioeconomic, environmental, and geographic variables and estimated logistic regressions to determine pixel-level CA determinants. These examples highlight the importance of investigating the pixel-scale causes of CA and the crucial role played by remote sensing technology.

In this section, we analyzed the relevance of remote sensing tools in investigating the causes of CA concerning geographic and socioeconomic aspects. We reviewed 61 peer-reviewed articles on the causes of CA and collated all driver factors (Table S2). Through an expert scoring method, the relevance between each factor and remote sensing was assessed (Fig. 5). In other words, we evaluated whether remote sensing tools and techniques can directly or indirectly provide strong evidence for monitoring the causes of CA.

The results showed that remote sensing tools play a dominant and advantageous role in assessing environmental indicators, with an average score of 3.6 (Fig. 5). Researchers often prefer to explain the degree of environmental impacts on CA using climatic, geophysical, and spatial information technology-related landscape patterns derived from remote sensing (Fig. 5). However, remote sensing tools show uncertainty (average score of 2.1) in assessing socioeconomic factors. Spatially characterized data tend to be more accessible through remote sensing compared to policy-related factors. For instance, distance variables, such as distance to hardened roads and market access, are easier to obtain using geospatial information technologies derived from remote sensing. Compiling all CA-related factors into a geographic information system (GIS) database often enables flexible and efficient access to site-specific information on CA determinants [97].

Geophysical factors

Geography is an important factor in the phenomenon of CA. Remote sensing technology plays a crucial role in capturing the environmental and topographical factors driving CA [97]. Geophysical information related to CA, deduced from remote sensing, reveals that in mountainous areas, the impact of environmental factors on CA is often more vital than socioeconomic variables [16]. In China's mountainous regions, characteristics such as the constraints of fine-grained cropland and steep slopes can explain 80% of the causes of CA [98]. Natural factors such as complex terrain, poor soils, and limited water resources frequently lead to the abandonment of cropland in mountainous and hilly areas [95]. Combining digital elevation models and geospatial accessibility analysis, Zhang et al. [98] found that CA was more likely to occur on steep slopes and plots far from laborers' shelters. Significant spatial correlations and strong statistical matches between CA rates,

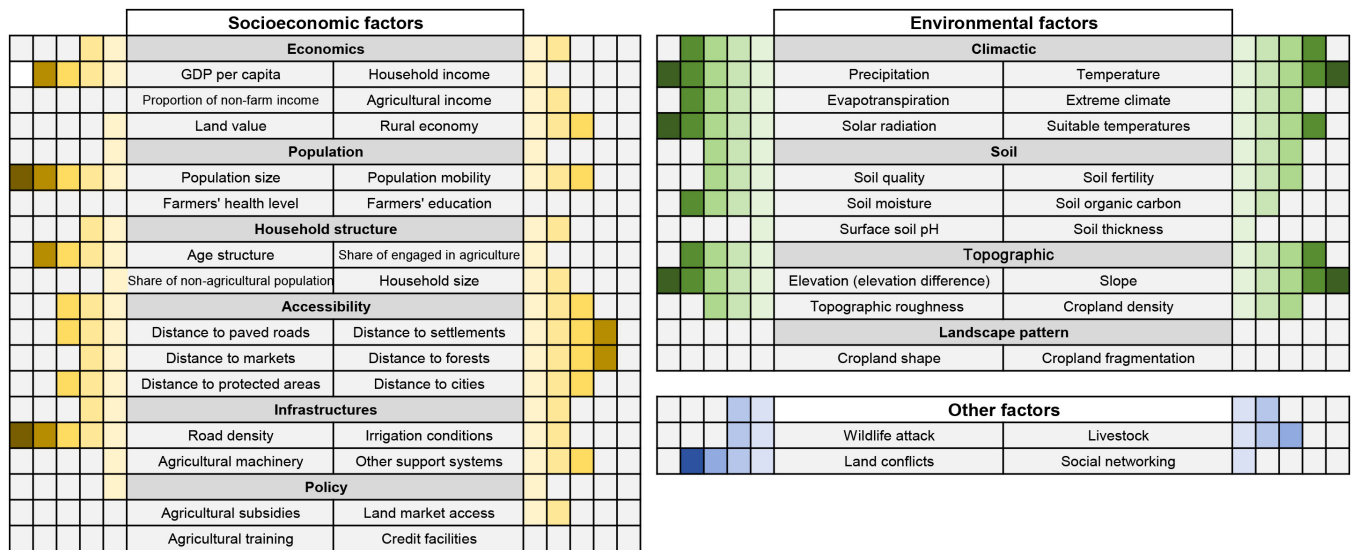


Fig. 5. The scoring map shows the relationship between remote sensing technology and the driving causes of CA. (Scores range from 0 to 5, indicating a transition from very uncorrelated to very strongly correlated. Here, we have collected 50 indicators commonly found in the previous 61 peer-reviewed articles [Table S2].)

elevation, and slope were also observed in Japan [99], Albania, and Romania [100].

Remote sensing imagery can provide detailed indirect information on climate, soil quality, plot characteristics, and natural hazards, thereby demonstrating the causal relationship between environmental factors and CA. Based on climatic factor inversions derived from remote sensing, Levers et al. [101] revealed that the primary pattern of agricultural abandonment in Europe is unfavorable climatic conditions for agriculture (i.e., low/high growing season degree days). In China's main grain-producing areas, low land quality is a crucial constraint on the long-term use of arable land [102]. Regarding landscape patterns portrayed by images, isolated and fragmented patches of farmland are more likely to be abandoned [103]. At fine-resolution image scales, different biophysical conditions can drive different abandonment dynamics, such as rainfed agriculture being more prone to abandonment than irrigated farmland [104].

Socioeconomic factors

Remote sensing technology also directly and indirectly reveals the impact of socioeconomic factors on CA. In high-resolution imagery, factors such as roads, urban sprawl, water sources, and settlements relevant to agricultural cultivation can be identified. Basic amenities such as irrigation and agricultural transport substantially influence farmers' decisions to sell their agricultural land. Cultivated land encroachment is observed in areas closely linked to transport networks and high economic activity levels [105]. Additionally, nighttime lights captured by remote sensing accurately reflect regional economic spatial patterns [106]. Li and Song [23] highlighted that socioeconomic factors in mountainous urban areas have a greater impact on the spatial distribution of abandoned land than environmental factors.

Remote sensing is also valuable for monitoring agricultural activities' intensity, such as changes in crop acreage and yields, thereby revealing the economic changes influencing CA [107]. Socioeconomic factors like population migration, agricultural mechanization levels, and changes in irrigation practices can be

indirectly assessed using remote sensing data [108]. Overlaying population raster data from different time points, Hou et al. [38] demonstrated the impact of population changes and urban–rural migration on CA. Based on Worldpop data, Zhang et al. [98] identified a nonlinear effect of agricultural labor age on CA.

Policy-oriented factors such as agricultural subsidies and technological and cultural systems pose challenges in quantifying their role using remote sensing techniques. In policy-driven regions, these factors often influence farmland owners' decisions to abandon their cropland [50]. Nonetheless, land-use changes, as direct outcomes of human activities, can partially reflect CA-related policy implementations and outcomes such as ecological restoration [47]. Zhang et al. [109] revealed the impact of ecological restoration projects on CA, highlighting the conversion of inefficient and highly sloped arable land into woodland or grassland under policies like the Grain-for-Green project. Yin et al. [110] found minimal re-cultivation after the Chechen war despite substantial subsidies, underscoring the profound and lasting impact of armed conflict on re-tillage.

Investigating the Ecological Effects Produced by CA Based on Remote Sensing

CA has multifaceted and wide-ranging ecological effects [52]. These effects can be either beneficial or detrimental, depending on conservation status, agro-climatic conditions, and local factors. On the positive side, CA can provide ecosystem services [111], such as increased carbon sequestration [32], a positive contribution to soil organic carbon [112], increased soil microbial activity [113], and soil and water conservation [114]. Some abandoned land naturally transforms into habitats [2,54], promoting biodiversity, especially among invertebrates [115], and improving landscape connectivity to valuable habitats for umbrella species [32].

However, abandoned land exacerbates climate change and poses a threat to global food security [38]. Vegetation regrowth increases weeds, pests, and pathogens in adjacent farmland [116], raising the incidence of harmful pests for cultivated crops

[9]. Moreover, uncontrolled accumulation of dry material on abandoned land fuels wildfires, escalating wildfire risks [117].

Remote sensing technology can provide large-scale, high temporal-spatial resolution terrestrial information critical for monitoring and assessing the ecological impacts of abandoned areas [64]. Using remotely sensed imagery, vegetation cover and growth in abandoned areas can be monitored to derive above-ground carbon stocks and dynamics [32]. Li et al. [118] inferred from remotely sensed net vegetation productivity data that abandoned croplands compensate for reduced net ecosystem productivity caused by urban expansion. Kong and Chen [119] found that abandonment of arable lands in Northwest China increased carbon stocks by 21.71 Tg C from 2000 to 2020. In contrast, in the European part of Russia, abandonment from 1990 to 2009 resulted in a net carbon sink of 470 Tg C, with substantial carbon sequestration occurring approximately 10 years after abandonment [120].

Remote sensing also monitors hydrology and evapotranspiration (ET) in abandoned areas. Combined with hydrological modeling, Yan et al. [121] found that seasonal abandonment increased groundwater depth by 4.5% and reduced daily runoff by 20.3%. Khorchani et al. [122] observed a 34% increase in ET due to secondary succession on abandoned surfaces, which subsequently increased water demand. Monitoring abandoned lands helps identify areas vulnerable to soil erosion and deforestation, particularly those with challenging terrain and poor soil conditions [56].

Among the negative environmental impacts, Nguyen et al. [123] linked CA in flood-prone areas to increased flood intensities using flood simulations and remote sensing imagery. Similarly, Dara et al. [124] mapped increased fire intensities in northern Kazakhstan due to cropland and pastureland abandonment using Landsat imagery. Changes in landscape structure and land use over time alter fire behavior parameters, increasing the risk of wildfires in successional vegetation [124,125]. Agricultural abandonment is a land use/land cover change contributing to increased wildfire risks, particularly at wildland-urban interfaces [126].

Investigating the ecological effects of CA using remote sensing is crucial for understanding the impact of large-scale human activities on natural environments. Reclaiming inefficient agricultural lands for active reforestation and ecological restoration can enhance ecological functions, including water regulation, soil restoration, nutrient cycling, and biodiversity promotion [48]. Based on geospatial modeling, Zheng et al. [2] found that the 101 Mha of globally abandoned agricultural land could mitigate 290 to 1,066 Mt CO₂ year⁻¹ in climate change, depending on land-use suitability and allocation strategies. Scenario analysis underscores the climate change mitigation potential of bioenergy plants on abandoned lands [86]. By leveraging comprehensive remote sensing data, a holistic understanding of the impacts of arable land abandonment on ecosystem structure, functions, and services can inform ecological protection and sustainable development policies.

Discussion

Challenges and uncertainties

Discrepancies in the definition of CA

Definitions of CA vary differently between studies and organizations, leading to uncertainty in identifying the extent and spatial-temporal patterns of abandoned lands. For instance,

Qiu et al. [35] defined the conversion of arable land to unused land as CA and found an annual CA rate of about 0.55% (approximately 862 km² year⁻¹) in China. In contrast, Zhang et al. [127] reported a CA rate of 18.59% in China, considering the conversion of arable land to noncultivated land (excluding cities and water bodies). The FAO defines abandoned farmland as land where farming activities have ceased for at least 2 to 5 years. However, this timeframe varies across countries due to different land resource management policies. From a land change perspective, CA refers to the process of transforming agricultural land into noncultivated uses. Jiang et al. [128] include farmland encroached upon by urban sprawl within the category of CA. These discrepancies highlight gaps that challenge the development and implementation of effective policies. Moreover, differences in crop rotation practices, land-use policies, and market conditions among countries further contribute to variations in the concept of CA.

Current CA research often fails to distinguish between active (retired cropland) and passive abandonment. Retired cropland refers to the intentional conversion of arable land into plantation forests, grasslands, or orchards under ecological restoration projects, reflecting a deliberate decision by landowners [45]. Hong et al. [47] used Landsat time series-based NDVI changes to differentiate between CA and intentional afforestation, and found that the proportion of intentional afforestation was greater than CA. Du et al. [45] found widespread agricultural CA and retirement in agro-pastoral zones. However, scenario studies suggest that deploying bioenergy crops in China's Grain for Green Project areas may be impractical [86].

Challenges in identifying abandoned land using remote sensing technology

The physicochemical characteristics of abandoned land exacerbate the challenge of identifying CA using remote sensing. During secondary succession, various vegetation types and surface features may coexist on abandoned lands, often with spectral characteristics similar to the surrounding natural vegetation (e.g., grasses and shrubs), leading to spectral mixing in remote sensing images [70]. In climatically favorable areas, pioneer plants such as herbs and shrubs can rapidly achieve vegetation cover on abandoned arable land, further complicating identification. This spectral mixing issue is particularly severe in lower-resolution images.

Mountainous areas, which have a high incidence of CA, present additional challenges due to their high slopes and fragmented parcels. Small, isolated areas and abandoned lands near natural forests are more difficult to observe using remote sensing. The complex terrain in these areas is prone to shadows and terrain occlusion, affecting image quality and making it difficult to clearly identify arable plots in low-resolution remote sensing imagery. Additionally, factors such as land cover diversity, topographic complexity, soil type, and humidity all impact the accuracy of CA monitoring. These factors affect the spectral characteristics and monitoring accuracy of remotely sensed images, necessitating comprehensive consideration in the monitoring and analysis process.

Remote sensing technology presents challenges for further research on CA. Despite the increasing availability of high-resolution remote sensing data, processing such imagery is complex and costly due to its large data volume. On a global scale, studies often rely on low-resolution but wide-coverage remote sensing data (e.g., MODIS), combined with global

land-use databases and models, to analyze the macro-distribution patterns and global change trends of abandoned lands [78]. At the regional scale, medium- and high-resolution remote sensing imagery, such as Landsat, Sentinel-2, and LiDAR, can be used to analyze distribution patterns [32]. Differences in data acquisition frequency and temporal coverage of various satellites also affect the ability to monitor continuously. Furthermore, there are limitations in data processing, model construction, and result interpretation.

Lack of potential assessment and scenario modeling of abandoned lands

A large number of studies have focused on identifying and understanding the drivers of abandoned lands, but there is a relative lack of assessment regarding the potential of these lands for ecological restoration, agricultural production, and sustainable use [29]. Innovative agroecological practices can be effective in rehabilitating degraded lands, which is crucial for enhancing the replanting potential of abandoned land [129]. Considering market access, Zheng et al. [2] concluded that global reuse of abandoned land could simultaneously provide 29 to 363 petacalories of food production potential and offer spatial indications for planting and afforestation. Jiang et al. [138] assessed the replanting potential of abandoned land in China's 3 main grain-producing regions, revealing that the maximum production potential of abandoned arable land could reach up to 8.5 million ton. However, pathways to reuse abandoned land are complex and often poorly understood [130].

In addition, the potential ecological impacts of CA need to be effectively assessed. Ecological restoration projects, such as China's Grain-to-Green Program, are among the most ecologically efficient means of mitigating climate change on abandoned land [131]. Considering natural revegetation in biodiversity priority areas and various measures, Gvein et al. [86] revealed that abandoned land could achieve a mitigation potential of 0.8 to 4.0 Gt CO₂ equivalents year⁻¹. The rewilding and afforestation of abandoned land could help mitigate global biodiversity loss and contribute to the creation of nature reserves [132], but these contributions still need to be quantitatively investigated. Furthermore, the potential impact of CA on ecosystem services such as soil structure, nutrient cycling, water retention, sand control, and soil erosion reduction has not yet been fully explored.

Future prospects

Clear definition of abandoned land

To improve the accuracy and consistency of research, there is an urgent need to reach a consensus on the definitional criteria of CA. Establishing a unified conceptual definition at both international and national levels is crucial. This definition should clarify the categorization of factors such as the reasons for abandonment, duration, nature of the land, and environmental impacts. Such a framework will help countries and regions to collect, analyze, and compare data more reliably, thereby improving the systematic nature and consistency of research [1].

There have been attempts to further categorize arable land abandonment. For example, in the Central Pyrenees of Spain, researchers have studied both active and passive land management [133]. In China, about 22.27% of the CA area is defined as intentional afforestation, which is larger the unused CA area [47]. Future research should not ignore ecological restoration efforts on efficient land, such as China's "Grain for Green" program

and the African Union's "Green Wall" program in the Sahelian zone of Africa [134]. It is worth mentioning that, in the process of identifying seasonal CA, it is necessary to provide a clearer definition of abandoned land to distinguish it from fallowing. Mixing fallowing with seasonal CA could introduce potential biases in cropland and abandonment statistics.

It is also essential to classify abandoned land based on different causes, such as economic reasons, natural disasters, and policy changes, and to distinguish between long-term and short-term abandonment. Such categorization will help better understand the mechanisms of land abandonment and its ecological, environmental, and socioeconomic impacts. This understanding will provide a foundation for developing targeted management strategies.

Integration of multiple data sources for identifying abandoned land

Field survey data can provide critical field validation and detailed information, particularly for complex terrains and areas that are challenging to capture with remote sensing imagery. With technological advancements and the emergence of cloud computing platforms for remote sensing, such as GEE, large-scale, high-precision mapping products for abandoned lands are becoming increasingly abundant [124]. The combined use of multiple data sources helps to compensate for the limitations of using a single data source.

Regarding satellite sensors, monitoring abandoned farmland requires balancing spatial and temporal resolution, as well as selecting appropriate sensor types and bands. Additionally, the complexity of land use and its spatiotemporal variations must be considered, necessitating the development of refined classification algorithms. Finally, integrating different types of remote sensing data with advanced data processing technologies can effectively improve the accuracy and timeliness of monitoring.

Currently, scholars are considering multisource data fusion methods to identify the spatial and temporal distribution of CA. For example, Paudel et al. [135] utilized a synergy of high-resolution remote sensing and social surveys to study CA in the Koshi River Basin, Central Himalayas. Some researchers have combined the high temporal resolution of MODIS data with the spatial resolution of other satellite sensors to achieve 30 m accuracy in identifying abandoned land [44,128]. Multisource data fusion analyses involving different remote sensing data types, such as optical, radar, and LiDAR, have also been conducted to improve CA identification [70].

Future research should focus on integrating ground observations, high-resolution remote sensing imagery, and socioeconomic data to more accurately analyze the spatial distribution, temporal changes, and influencing factors of abandoned land. Rapid processing and real-time monitoring of large-scale abandoned cropland spatial information data can provide scientific support for sustainable land management and resource conservation.

Modeling temporal-spatial pattern of CA risks and future scenarios

The study estimates that the global area of abandoned land is likely to continue increasing, particularly in Asia and Africa [136]. Future research prospects on CA lie in strengthening our ability to understand, model, and predict the spatial and temporal trends of abandoned cropland. Establishing a high-precision

monitoring and assessment system for abandoned land will be crucial. This system should integrate remote sensing technology, GIS, the Internet of Things, and ground survey data.

The Land Use-based Integrated Sustainability Assessment territorial modeling platform from the Joint Research Centre of the European Commission, which focuses on regional and local projections of future land abandonment between 2015 and 2030 in the European Union, is an example of such an effort [137]. To address the lag in mapping and observing CA [138], there is an urgent need to predict future high-risk areas for CA as well as temporal and spatial trends.

Moreover, few studies have discussed possible ways to reuse abandoned farmland [138]. The collapse of the Soviet Union has created an enormous, yet currently untapped, potential for carbon sequestration, offering substantial opportunities for offsetting industrial carbon emissions and reforestation [32]. Additionally, the potential for food production from re-cultivation on abandoned land remains largely unknown. Therefore, timely information on the extent of arable land abandonment and its future trends is urgently needed to assess the trade-offs between agricultural development and the provision of environmental services [138].

Future research should focus on simulating and predicting the maximum potential use of abandoned land by constructing temporal and spatial dynamic models and conducting scenario analyses. This approach will provide policymakers with a scientific foundation and effective response strategies for managing abandoned lands.

The “cause–pattern–effect” trinity system framework

In this work, we proposed an observation framework for abandoned land that couple causes, patterns, and effects (Fig. 6). This is a comprehensive framework that uses remote sensing technology to reveal the spatiotemporal patterns, driving factors, and ecological benefits of CA. The “cause–pattern–effect” trinity system framework provides multidimensional information for abandoned farmland surveys, further offering robust insights for policymakers, such as the nonlinear and cascading effects of drivers and ecological benefits on CA. In terms of its causes, spatial data on economic, social, policy, and natural factors can be used to develop a geospatial database (Fig. 6). This database allows for the analysis of the formation mechanisms of arable CA in different regions and for various types of land. The analysis includes the impact of economic development, changes in agricultural policies on land-use decisions [139], and the effects of natural disasters, land quality, and ecological factors.

An accurate system for identifying and monitoring abandoned land is established using high-resolution satellite imagery, drones, and ground observation data (Fig. 6). This multiscale detection and integration approach captures the spatial distribution and temporal changes of abandoned land, providing detailed information on the boundaries, area, and morphology of the abandoned plots. This enables a thorough understanding of the patterns of CA (Fig. 6).

The framework also focuses on the ecological effects of abandoned land using remote sensing technology (Fig. 6). Key

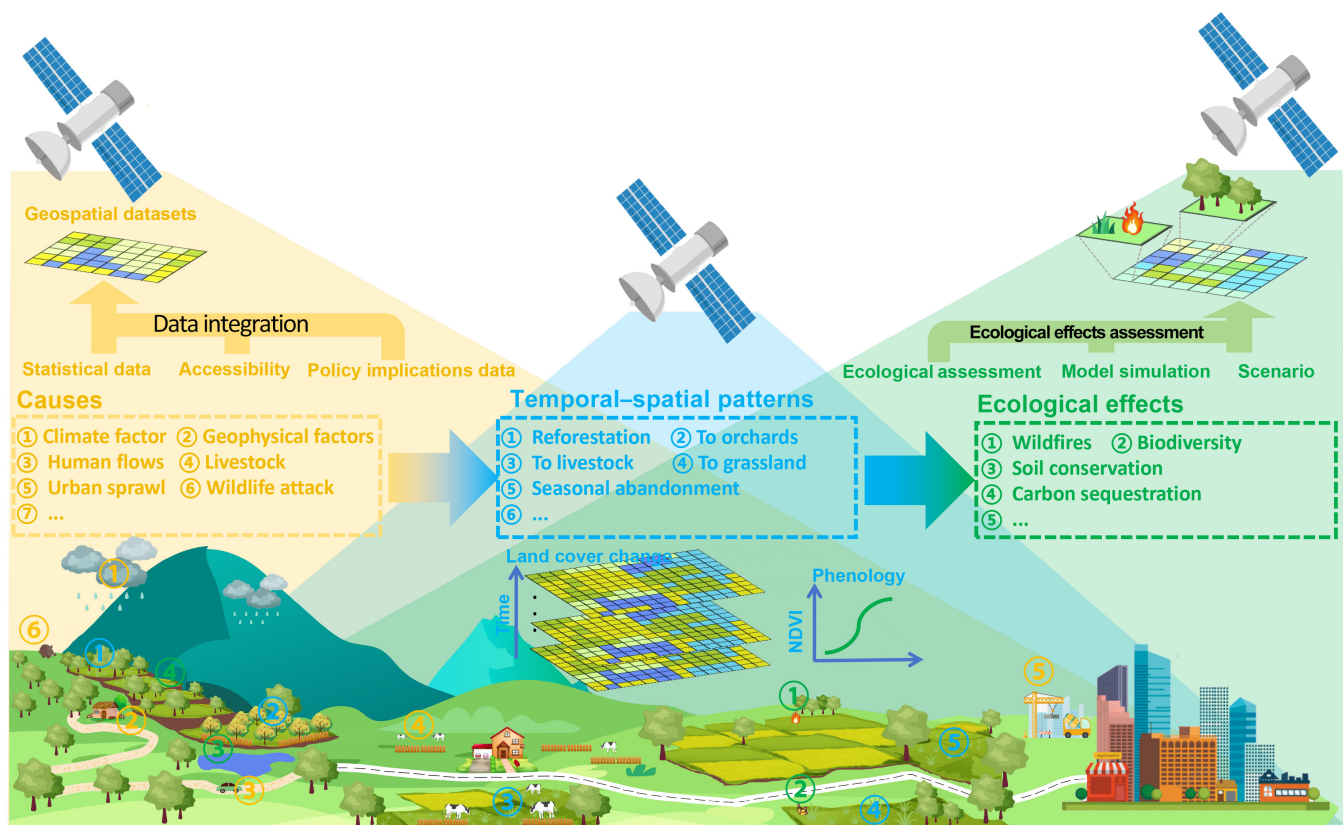


Fig. 6. The conceptual framework couples the causes, patterns, and impacts of CA based on remote sensing images and technologies. (Remote sensing technology bridges the “cause–location–ecological effect” pillars of CA. By integrating multisource data and fostering interdisciplinary cooperation, this approach provides scientific support for sustainable abandonment management and resource use. A comprehensive analysis of these 3 dimensions offers an entry point for policy action to promote the optimal use of land resources and sustainable development.)

aspects include carbon fixation, soil and water conservation, and biodiversity restoration. Remote sensing, combined with ecological modeling, quantifies the value of ecological services and their contribution to biodiversity [80]. Additionally, the framework monitors unfavorable risk factors, such as wildfires, biological invasions, and arthropod colonization. The ultimate goal is to quantify the potential impacts of abandonment on ecosystems and provide a scientific basis for developing restoration strategies and policies.

Conclusion

In this work, we systematically reviewed the crucial role played by remote sensing methodologies in revealing the causes, spatiotemporal distribution, and ecological effects of CA. Based on this, we discussed the current challenges in CA research and future priorities. The bibliometric analysis shows that most case studies on CA monitoring are conducted in Europe and Asia. Among these studies, Landsat and MODIS satellites are used in over half (52.63%) of cases. Chronologically, high CA rates were observed in the EU and the US before 2000, and in China between 2010 and 2020, and are projected to increase in Africa in the future. Remote sensing-based methods for monitoring abandoned land broadly include 2 main approaches: spectral feature classification and time-dynamic trajectory. Among the 8 subcategories, vegetation phenology methods have the highest average overall accuracy at 89.33%. In addition, remote sensing and geospatial analyses play a crucial role in monitoring the causes and ecological effects of CA, including food loss, wildfire risk, ecosystem services, and biodiversity. Overall, we presented 3 major current challenges and 4 future perspectives. Importantly, the “cause–pattern–effects” trinity framework we proposed is essential for integrating CA research. Our review provided a systematic perspective for the integration of CA research and policy practice.

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Data Availability

All data used in this study, including Tables S1 to S3, are publicly available at <https://doi.org/10.6084/m9.figshare.28039832>.

Supplementary Materials

Tables S1 to S3

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